

Lab 2 — Agile Backlog & Sprint Simulation

Smart Lab Equipment & Slot Reservation Portal · PES University, Dept. of CSE · Problem Statement #01

Jira project **Smart Lab Portal** (key SLP), Scrum, company-managed. Backlog derived from the Lab 1 requirements table.

1. Epics and User Stories

Seven Epics covering the five functional requirements, both nonfunctional requirements and use case UC-13 from Lab 1. Twenty-four User Stories, 127 story points, estimated on the Fibonacci scale.

Epic 1: Slot Reservation & Availability

Traces to FR-001

Lets an authenticated student see live instrument availability and hold a slot of at most 2 hours, up to 7 days ahead, on an instrument whose calibration is valid.

Story	User story	Priority	Points	Sprint
1.1 Real-Time Availability View	As a student, I want to see live availability for every instrument across the next 7 days, so that I can pick a free slot without walking to the lab.	High	5	Sprint 1
1.2 Reserve a Slot	As a student, I want to reserve a slot of up to 2 hours on a calibrated instrument, so that the instrument is guaranteed free when I arrive.	High	8	Sprint 1
1.3 Booking-Rule Enforcement	As a lab administrator, I want to have requests over 120 minutes or more than 7 days ahead rejected, so that no one hoards scarce hardware.	High	3	Sprint 1
1.4 Reservation Token Issue	As a student, I want to receive a unique reservation token on confirmation, so that I can prove my booking at the issue counter.	High	3	Sprint 1

4 stories, 19 story points.

Epic 2: Calibration Governance

Traces to FR-002

Makes calibration state gate bookability rather than merely be displayed, so no reading is ever taken on an out-of-calibration instrument.

Story	User story	Priority	Points	Sprint
2.1 Record Calibration Event	As a lab technician, I want to record date performed, next-due date and certificate ID against an instrument, so that every instrument's calibration state is on record.	High	5	Sprint 1
2.2 Auto-Withdraw Expired Instruments	As a faculty supervisor, I want to have an instrument leave the bookable inventory the moment its next-due date passes, so that no reading is taken on an out-of-calibration instrument.	High	8	Sprint 1
2.3 Notify Affected Reservation Holders	As a student, I want to be told within 5 minutes if my future booking is voided by calibration expiry, so that I can rebook on another instrument in time.	Medium	5	Backlog

3 stories, 18 story points.

Epic 3: Equipment Custody

Traces to FR-003

Ties physical custody of an instrument to the booking record through token validation and server-side check-out / check-in timestamps.

Story	User story	Priority	Points	Sprint
3.1 Validate Token at Issue	As a lab technician, I want to validate the student's reservation token before handing over an instrument, so that only the rightful booker takes it.	High	5	Sprint 2
3.2 Reject Invalid Tokens	As a lab technician, I want to see expired, reused or wrong-instrument tokens rejected with a clear reason, so that I can refuse issue confidently at the counter.	High	3	Sprint 2
3.3 Server-Side Custody Timestamps	As a auditor, I want to have check-out and check-in timestamps stored server-side against the reservation, so that a late return can be proven from the record.	High	5	Sprint 2

3 stories, 13 story points.

Epic 4: Late-Return Penalties

Traces to FR-004

Implements the disciplinary rule: overdue computation, a 30-minute grace window, strikes, and suspension on the third strike inside a rolling 30 days.

Story	User story	Priority	Points	Sprint
4.1 Compute Overdue at Check-In	As a lab technician, I want to have overdue duration computed automatically at check-in, so that I never have to judge lateness by hand.	High	5	Sprint 2
4.2 Grace Window and Strike	As a student, I want to have a 30-minute grace window before a strike is recorded, so that a queue at the issue counter does not penalise me.	High	5	Sprint 2
4.3 Third-Strike Suspension	As a faculty supervisor, I want to have booking privileges suspended for 14 days on a third strike in 30 days, so that repeat offenders stop blocking lab capacity.	High	8	Sprint 2
4.4 Penalty Notifications	As a student, I want to be notified within a minute of a strike or a suspension, so that I am never surprised by a blocked booking.	Medium	3	Backlog

4 stories, 21 story points.

Epic 5: Cancellation & Waitlist

Traces to FR-005

Recovers scarce lab capacity from abandoned bookings instead of leaving instruments idle.

Story	User story	Priority	Points	Sprint
5.1 Cancel a Reservation	As a student, I want to cancel a reservation up to 2 hours before its start time, so that I am not penalised when my plans change.	Medium	3	Backlog
5.2 Join a Waitlist	As a student, I want to join the waitlist for a fully booked instrument-slot, so that I get a chance if someone cancels.	Medium	5	Backlog
5.3 Sequential Waitlist Offer	As a waitlisted student, I want to be offered a released slot in waitlist order with 30 minutes to accept, so that the offer is fair and does not stall on one person.	Medium	8	Backlog

5.4 No-Show Marking	As a faculty supervisor, I want to see unused, uncanceled reservations marked NO_SHOW, so that ghost bookings are visible in utilisation data.	Low	3	Backlog
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4 stories, 19 story points.

Epic 6: Access Control, Audit & Concurrency

Traces to NFR-001, NFR-002

Platform integrity: role-based access, an append-only evidence trail, and correct isolation on the contended reservation write.

Story	User story	Priority	Points	Sprint
6.1 Role-Based Access Control	As a lab technician, I want to have calibration and penalty actions restricted to my role, so that students cannot alter their own records.	High	5	Sprint 1
6.2 Append-Only Audit Log	As a auditor, I want to have every reservation, calibration, custody and penalty event logged immutably with actor, action, timestamp and before/after state, so that a suspension can be contested with evidence.	High	8	Sprint 2
6.3 Twelve-Month Audit Export	As a lab technician, I want to export 12 months of the audit log as CSV, so that I can answer an external audit without developer help.	Low	3	Backlog
6.4 Race-Free Concurrent Booking	As a student, I want to have exactly one of many simultaneous requests for the same slot succeed, within 200 ms, so that double-booking is impossible rather than merely unlikely.	High	13	Backlog

4 stories, 29 story points.

Epic 7: Lab Utilisation Reporting

Traces to UC-13

Gives the Faculty Supervisor visibility of how lab capacity is actually used.

Story	User story	Priority	Points	Sprint
7.1 Utilisation Report	As a faculty supervisor, I want to see booked versus actually used hours per instrument, so that I can find where lab capacity is being wasted.	Medium	5	Backlog
7.2 Filter Utilisation by Period	As a faculty supervisor, I want to filter utilisation by date range and department, so that I can justify new equipment purchases with evidence.	Low	3	Backlog

2 stories, 8 story points.

2. Sprint results

Sprint	Started	Items	Points	Completed
SLP Sprint 1	02/Sep/26 12:11 PM	7	37	7 of 7
SLP Sprint 2	02/Sep/26 12:24 PM	7	39	7 of 7
Remaining backlog	—	10	51	—

Sprint 1 goal. A student can find and hold a slot on an instrument that is provably in calibration, and only a technician can change calibration data.

Sprint 2 goal. An instrument can only be issued against a valid token, and a late return produces a strike that is defensible from the audit log.

Both sprints were time-boxed to one week ending 09/Sep/26. Epics 1–4 closed completely, Epic 6 partially (13 of 29 points), Epics 5 and 7 untouched.

The screenshot shows the Jira Smart Lab Portal interface. The left sidebar contains navigation options like 'For you', 'Recent', 'Starred', 'Apps', 'Plans', 'Spaces', 'Teams', and 'Customize sidebar'. The main area is titled 'SLP board' and shows a 'Backlog' view. A 'Plan your sprint' section at the top right of the main area contains a message: 'Plan your sprint. Drag work items from the Backlog section or create new ones to plan the work for this sprint. Select Start sprint when you're ready.' Below this, the 'Backlog (10 work items)' section lists 10 items, each with a checkbox, a story key (e.g., SLP-22), a story title (e.g., Story 2.3: Notify Affected Reservation Holders), an epic link (e.g., Epic 2: Calibration ...), a status (e.g., To Do), and a point value (e.g., 5). The items are sorted by point value in descending order. The bottom of the screen shows a 'Create' button.

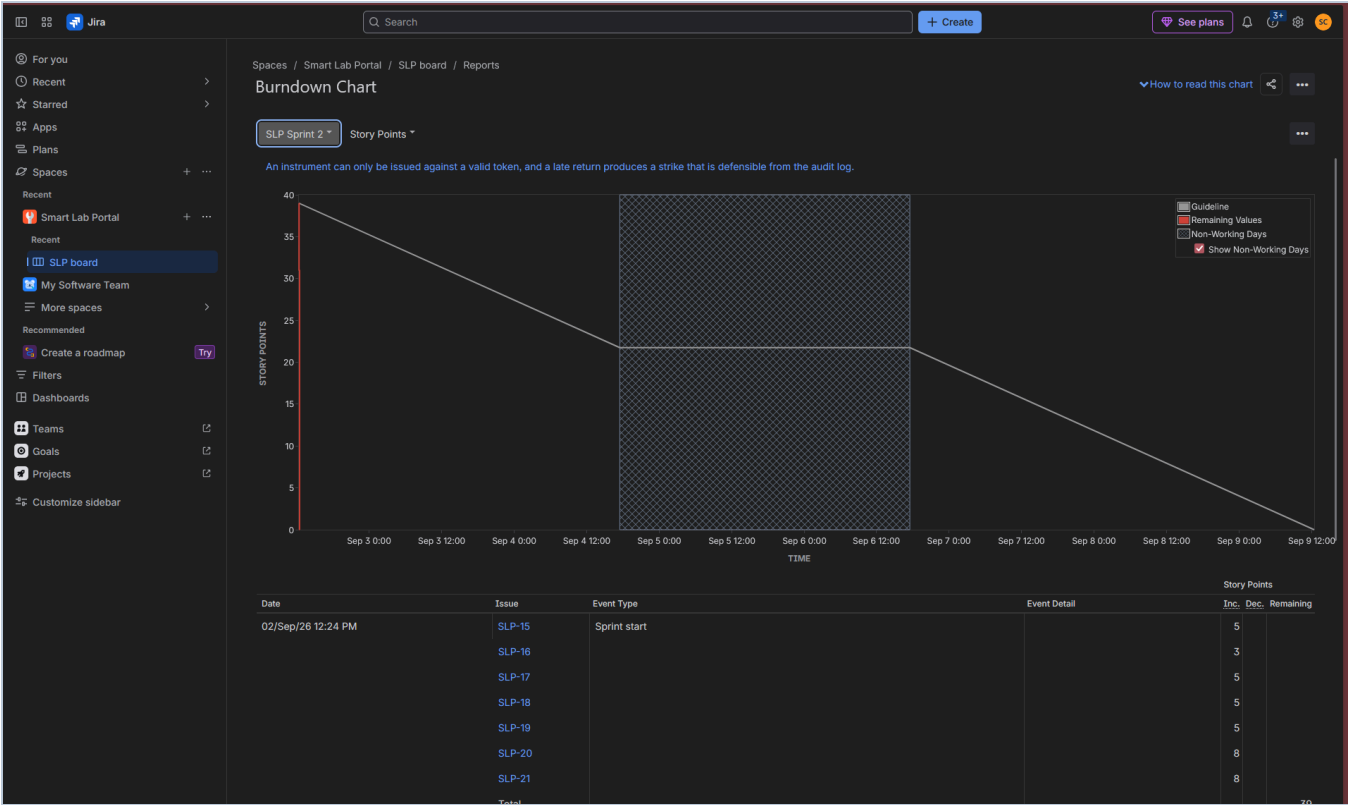
3. Burndown charts

In both charts the Remaining Values line falls from the committed total to zero within minutes of the sprint starting and then runs flat along the zero axis for the rest of the one-week time box, while the guideline descends slowly to meet it on 09/Sep. The line sits below the guideline not because the team was fast but because seven days of planned work were transacted in about ten minutes of elapsed time, so neither chart supports a velocity claim. Read as a real sprint, the shape is the signature of severe under-commitment.

SLP Sprint 1 – 37 story points



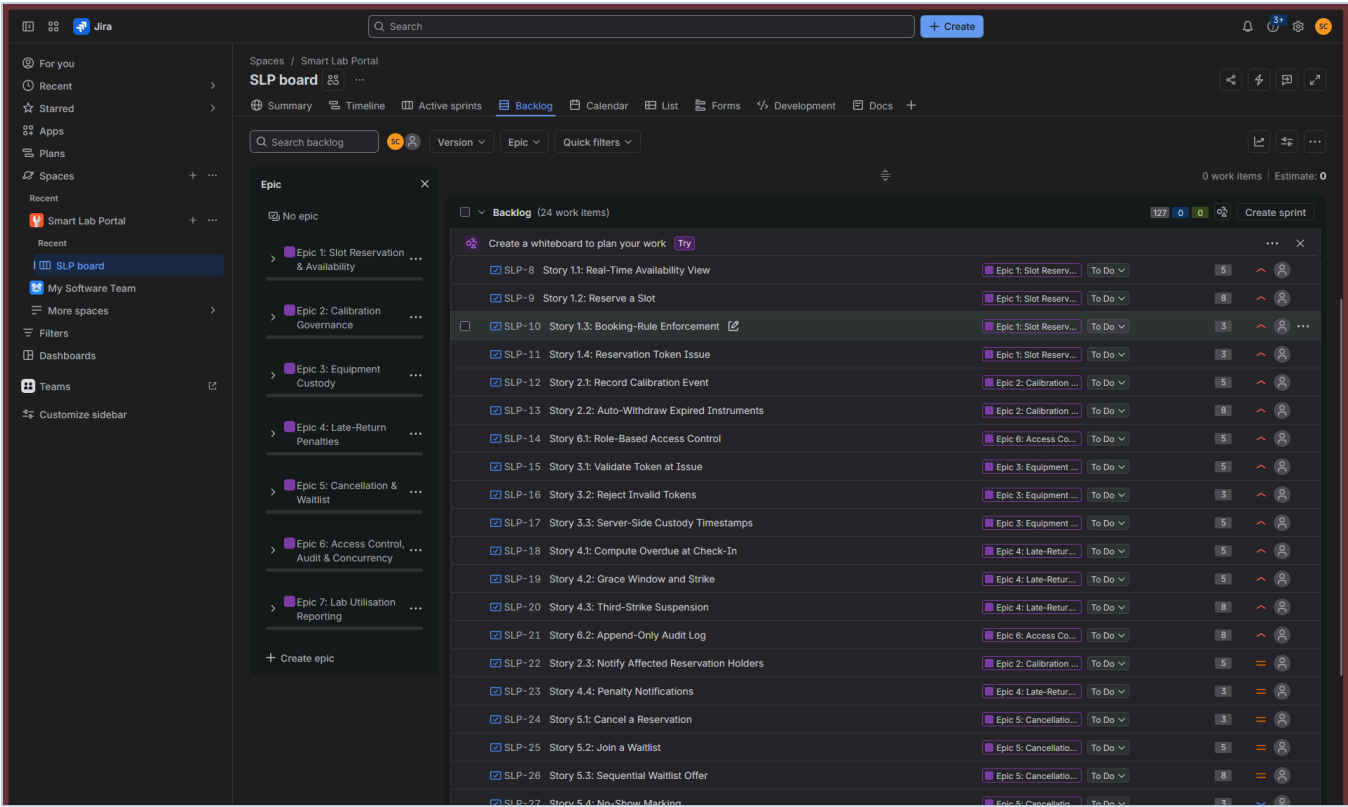
SLP Sprint 2 — 39 story points



4. Jira workspace

Backlog with Epics and User Stories

Epic panel showing all seven Epics, 24 User Stories in the backlog, 127 story points estimated.



Story point assignments

Sprint 2 loaded with 7 items totalling 39 points. Story points and priority icons on each row, Epic chip on every story, 51 points left in the backlog below.

The screenshot shows the Jira interface for the 'SLP board'. The left sidebar contains navigation options like 'For you', 'Recent', 'Starred', 'Apps', 'Plans', 'Spaces', 'Recent', 'Smart Lab Portal', 'My Software Team', 'More spaces', 'Filters', 'Dashboards', 'Teams', and 'Customize sidebar'. The main area displays the 'SLP board' with tabs for 'Summary', 'Timeline', 'Active sprints', 'Backlog', 'Calendar', 'List', 'Forms', 'Development', and 'Docs'. The 'Backlog' tab is active, showing a list of work items. The 'SLP Sprint 2' (9 Sep - 16 Sep, 7 work items) is visible, with a total of 39 points. The backlog shows 10 work items with a total of 51 points. Each work item is a story card with a title, epic chip, status, points, and priority icon.

Story ID	Story Title	Epic	Status	Points	Priority
SLP-15	Story 3.1: Validate Token at Issue	Epic 3: Equipment...	To Do	5	High
SLP-16	Story 3.2: Reject Invalid Tokens	Epic 3: Equipment...	To Do	3	High
SLP-17	Story 3.3: Server-Side Custody Timestamps	Epic 3: Equipment...	To Do	5	High
SLP-18	Story 4.1: Compute Overdue at Check-In	Epic 4: Late-Retur...	To Do	5	High
SLP-19	Story 4.2: Grace Window and Strike	Epic 4: Late-Retur...	To Do	5	High
SLP-20	Story 4.3: Third-Strike Suspension	Epic 4: Late-Retur...	To Do	8	High
SLP-21	Story 6.2: Append-Only Audit Log	Epic 6: Access Co...	To Do	8	High
SLP-22	Story 2.3: Notify Affected Reservation Holders	Epic 2: Calibration...	To Do	5	High
SLP-23	Story 4.4: Penalty Notifications	Epic 4: Late-Retur...	To Do	3	High
SLP-24	Story 5.1: Cancel a Reservation	Epic 5: Cancellatio...	To Do	3	High
SLP-25	Story 5.2: Join a Waitlist	Epic 5: Cancellatio...	To Do	5	High
SLP-26	Story 5.3: Sequential Waitlist Offer	Epic 5: Cancellatio...	To Do	8	High
SLP-27	Story 5.4: No-Show Marking	Epic 5: Cancellatio...	To Do	3	High
SLP-28	Story 6.3: Twelve-Month Audit Export	Epic 6: Access Co...	To Do	3	High
SLP-29	Story 6.4: Race-Free Concurrent Booking	Epic 6: Access Co...	To Do	13	High
SLP-30	Story 7.1: Utilisation Report	Epic 7: Lab Utilisat...	To Do	5	High
SLP-31	Story 7.2: Filter Utilisation by Period	Epic 7: Lab Utilisat...	To Do	3	High

Sprint board — SLP Sprint 1

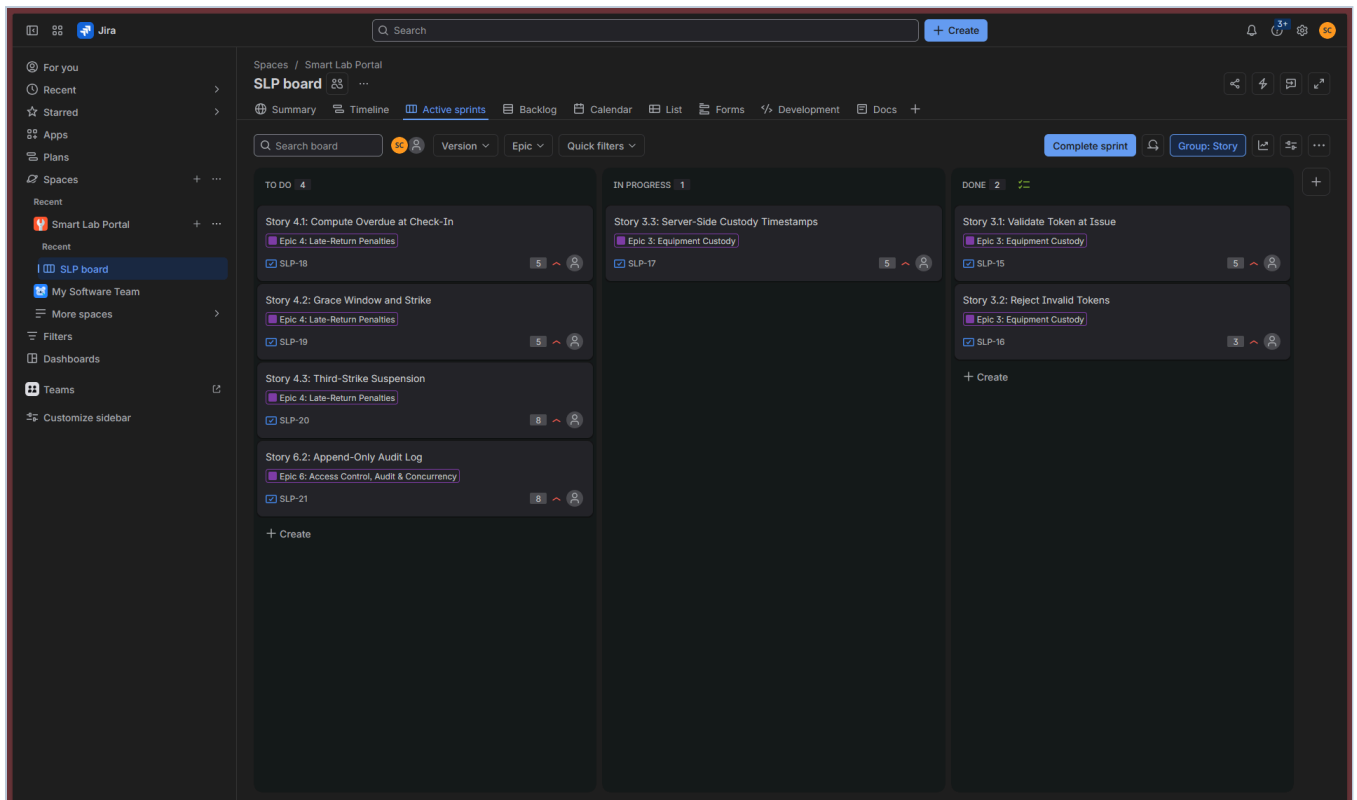
Active sprint view mid-sprint: 2 To Do, 2 In Progress, 3 Done.

The screenshot shows the Jira interface for the 'SLP board' in the 'Active sprints' view. The left sidebar is the same as the previous screenshot. The main area displays the 'SLP board' with tabs for 'Summary', 'Timeline', 'Active sprints', 'Backlog', 'Calendar', 'List', 'Forms', 'Development', and 'Docs'. The 'Active sprints' tab is active, showing a Kanban board with three columns: 'TO DO' (2 items), 'IN PROGRESS' (2 items), and 'DONE' (3 items). Each item is a story card with a title, epic chip, status, points, and priority icon.

Column	Story ID	Story Title	Epic	Status	Points	Priority
TO DO	SLP-13	Story 2.2: Auto-Withdraw Expired Instruments	Epic 2: Calibration Governance	To Do	8	High
	SLP-14	Story 6.1: Role-Based Access Control	Epic 6: Access Control, Audit & Concurrency	To Do	5	High
IN PROGRESS	SLP-11	Story 1.4: Reservation Token Issue	Epic 1: Slot Reservation & Availability	In Progress	3	High
	SLP-12	Story 2.1: Record Calibration Event	Epic 2: Calibration Governance	In Progress	5	High
DONE	SLP-8	Story 1.1: Real-Time Availability View	Epic 1: Slot Reservation & Availability	Done	5	High
	SLP-9	Story 1.2: Reserve a Slot	Epic 1: Slot Reservation & Availability	Done	8	High
	SLP-10	Story 1.3: Booking-Rule Enforcement	Epic 1: Slot Reservation & Availability	Done	3	High

Sprint board — SLP Sprint 2

Active sprint view mid-sprint: 4 To Do, 1 In Progress, 2 Done.



Sprint completion

